PUBLISHED: 16-DEC-2019
2020.0 F-TYPE (X152), 100-00

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX: SUSPENSION CONTROL MODULE (SUMB) (G1809517)

DESCRIPTION AND OPERATION

INTEGRATED SUSPENSION CONTROL MODULE (ISCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- If the control module or a component is at fault or may beat fault and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If diagnostic trouble codes are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed through the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all diagnostic trouble codes (DTCs) that could be set in the integrated suspension control module, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: Vehicle Dynamic Suspension (204-05 Vehicle Dynamic Suspension, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C101D-11	Left Front Vertical Acceleration Sensor - Circuit short to ground	▪ Front left vertical acceleration sensor circuit short circuit to ground ▪ Vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check front left vertical acceleration sensor circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101D-14	Left Front vertical Acceleration Sensor - Short to ground or open	▪ Front left vertical acceleration sensor circuit short to ground or open circuit ▪ Vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check the front left vertical acceleration sensor circuit for short circuit to ground, open circuit. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101D-1C	Left Front Vertical Acceleration Sensor - Circuit voltage out of range	▪ Front left vertical acceleration sensor circuit short circuit to other circuit or short circuit to power, open circuit, high resistance ▪ Vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check front left vertical acceleration sensor circuit for short circuit to another circuit, short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101D-22	Left Front Vertical Acceleration Sensor - Signal amplitude > maximum	▪ Front left vertical acceleration sensor signal amplitude above maximum ▪ Front left vertical acceleration sensor signal circuit short circuit to another circuit ▪ Front left vertical acceleration sensor insecurely mounted ▪ Front left vertical acceleration sensor fault	▪ With vehicle parked on a level surface, read front left vertical accelerometer voltage and check it lies in range 1.9 to 2.1 volts. If voltage values are outside this range, check front left vertical acceleration sensor signal circuit for short circuit to another circuit, loose connections and repair as required ▪ If no wiring faults are present, check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101D-26	Left Front Vertical Acceleration Sensor - Signal rate of change below threshold	▪ Front left vertical acceleration sensor signal not changing ▪ Front left vertical acceleration sensor signal circuit short circuit to another circuit ▪ Front left vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check front left vertical accelerometer signal circuit for short circuit to another circuit and repair as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101D-78	Left Front Vertical Acceleration Sensor - Alignment or adjustment incorrect	▪ Front left vertical acceleration sensor alignment or adjustment incorrect ▪ Front left vertical acceleration sensor bracket bent ▪ Front left vertical acceleration sensor damaged	▪ Check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ Check the integrity and mounting of the sensor bracket and secure or replace as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101E-11	Right Front Vertical Acceleration Sensor - Circuit short to ground	▪ Front right vertical acceleration sensor circuit short circuit to ground ▪ Front right vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check front right vertical acceleration sensor circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C101E-1C	Right Front Vertical Acceleration Sensor - Circuit voltage out of range	▪ Front right vertical acceleration sensor circuit short circuit to other circuit or short circuit to power, open circuit, high resistance ▪ Front right vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check front right vertical acceleration sensor circuit for short circuit to other circuit or short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101E-22	Right Front Vertical Acceleration Sensor - Signal amplitude > maximum	▪ Front right vertical acceleration sensor signal amplitude above maximum ▪ Front right vertical acceleration sensor signal circuit short circuit to another circuit ▪ Front right vertical acceleration sensor insecurely mounted ▪ Front right vertical acceleration sensor fault	▪ With vehicle parked on a level surface, read front right vertical accelerometer voltage and check it lies in range 1.9 to 2.1 volts. If voltage values are outside this range, check front right vertical acceleration sensor signal circuit for short circuit to another circuit, loose connections and repair as required ▪ If no wiring faults are present, check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101E-26	Right Front Vertical Acceleration Sensor - Signal rate of change below threshold	▪ Front right vertical acceleration sensor signal not changing ▪ Front right vertical acceleration sensor signal circuit short circuit to another circuit ▪ Front right vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check front right vertical accelerometer signal circuit for short circuit to another circuit and repair as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C101E-78	Right Front Vertical Acceleration Sensor - Alignment or adjustment incorrect	▪ Front right vertical acceleration sensor alignment or adjustment incorrect ▪ Front right vertical acceleration sensor bracket bent ▪ Front right vertical acceleration sensor damaged	▪ Check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ Check the integrity and mounting of the sensor bracket and secure or replace as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C1024-00	System Temporarily Disabled Due To Power Interruption During Driving - No sub type information	▪ Loss of power to control module when driving	▪ Refer to the electrical circuit diagrams and check power and ground circuits to the integrated suspension control module for intermittent or poor connections. Repair circuits as required, Clear the DTCs and retest
C102C-11	Right Rear Vertical Acceleration Sensor - Circuit short to ground	▪ Rear right vertical acceleration sensor circuit short circuit to ground ▪ Rear right vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check rear right vertical acceleration sensor circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest ▪ If no wiring faults are present, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C102C-14	Right Rear Vertical Acceleration Sensor - Circuit short to ground or open	▪ Rear right vertical acceleration sensor circuit short circuit to ground or open circuit ▪ Vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check the rear right vertical acceleration sensor circuit for short circuit to ground, open circuit. Repair circuit as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C102C-1C	Right Rear Vertical Acceleration Sensor - Circuit voltage out of range	▪ Rear right vertical acceleration sensor circuit partial short circuit to other circuit or short circuit to power, open circuit, high resistance ▪ Rear right vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check rear right vertical acceleration sensor circuit for partial short circuit to other circuit or short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ If no wiring faults are present, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C102C-22	Right Rear Vertical Acceleration Sensor - Signal amplitude > maximum	▪ Rear right vertical acceleration sensor signal amplitude above maximum ▪ Rear right vertical acceleration sensor signal circuit short circuit to another circuit ▪ Rear right vertical acceleration sensor insecurely mounted ▪ Rear right vertical acceleration sensor fault	▪ With vehicle parked on a level surface, read rear right vertical accelerometer voltage and check it lies in range 1.9 to 2.1 volts. If voltage values are outside this range, check rear right vertical acceleration sensor signal circuit for short circuit to another circuit, loose connections and repair as required ▪ If no wiring faults are present, check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new sensor as required. Clear the DTCs and retest
C102C-26	Right Rear Vertical Acceleration Sensor - Signal rate of change below threshold	▪ Rear right vertical acceleration sensor signal not changing ▪ Rear right vertical acceleration sensor signal circuit short circuit to another circuit ▪ Rear right vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check rear right vertical accelerometer signal circuit for short circuit to another circuit and repair as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new sensor as required. Clear the DTCs and retest
C102C-78	Right Rear Vertical Acceleration Sensor - Alignment or adjustment incorrect	▪ Rear right vertical acceleration sensor alignment or adjustment incorrect ▪ Rear right vertical acceleration sensor bracket bent ▪ Rear right vertical acceleration sensor damaged	▪ Check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ Check the integrity and mounting of the sensor bracket and secure or replace as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new sensor as required. Clear the DTCs and retest
C1030-11	Left Rear Vertical Acceleration Sensor - Circuit short to ground	▪ Rear left vertical acceleration sensor circuit short circuit to ground ▪ Rear left vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check rear left vertical acceleration sensor circuit for short circuit to ground. Repair circuit as required. Clear the DTCs and retest ▪ If no wiring faults are present, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest
C1030-1C	Left Rear Vertical Acceleration Sensor - Circuit voltage out of range	▪ Rear left vertical acceleration sensor circuit partial short circuit to other circuit or short circuit to power, open circuit, high resistance ▪ Rear left vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check rear left vertical acceleration sensor circuit for partial short circuit to other circuit or short circuit to power, open circuit, high resistance. Repair circuit as required. Clear the DTCs and retest ▪ If no wiring faults are present, check and install a new vertical acceleration sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1030-22	Left Rear Vertical Acceleration Sensor - Signal amplitude > maximum	▪ Rear left vertical acceleration sensor signal amplitude above maximum ▪ Rear left vertical acceleration sensor signal circuit short circuit to another circuit ▪ Rear left vertical acceleration sensor insecurely mounted ▪ Rear left vertical acceleration sensor fault	▪ With vehicle parked on a level surface, read rear left vertical accelerometer voltage and check it lies in range 1.9 to 2.1 volts. If voltage values are outside this range, check rear left vertical acceleration sensor signal circuit for short circuit to another circuit, loose connections and repair as required ▪ If no wiring faults are present, check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new sensor as required. Clear the DTCs and retest
C1030-26	Left Rear Vertical Acceleration Sensor - Signal rate of change below threshold	▪ Rear left vertical acceleration sensor signal not changing ▪ Rear left vertical acceleration sensor signal circuit short circuit to another circuit ▪ Rear left vertical acceleration sensor fault	▪ Refer to the electrical circuit diagrams and check rear left vertical accelerometer signal circuit for short circuit to another circuit and repair as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new sensor as required. Clear the DTCs and retest
C1030-78	Left Rear Vertical Acceleration Sensor - Alignment or adjustment incorrect	▪ Rear left vertical acceleration sensor alignment or adjustment incorrect ▪ Rear left vertical acceleration sensor bracket bent ▪ Rear left vertical acceleration sensor damaged	▪ Check the sensor is correctly mounted and secure the sensor as required. Clear the DTCs and retest ▪ Check the integrity and mounting of the sensor bracket and secure or replace as required. Clear the DTCs and retest ▪ If the fault persists, check and install a new sensor as required. Clear the DTCs and retest
C1056-66	Left Rear Air Spring Air Supply - Signal has too many transitions/events	▪ Harness fault short circuit to ground, open circuit high resistance ▪ Valve block not connected ▪ Valve block electrical fault short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module circuit to ground, open circuit, high resistance ▪ Check the connection of the valve block ▪ Refer to the electrical circuit diagrams and check the valve block for short circuit to ground, short circuit to power, open circuit, high resistance
C1058-66	Right Rear Air Spring Air Supply - Signal has too many transitions/events	▪ Harness fault short circuit to ground, open circuit high resistance ▪ Valve block not connected ▪ Valve block electrical short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module circuit to ground, open circuit, high resistance ▪ Check the connection of the valve block ▪ Refer to the electrical circuit diagrams and check the valve block for short circuit to ground, short circuit to power, open circuit, high resistance
C105A-01	Actuator Circuit Group A - General electrical failure	**NOTE:** Circuit reference - VBATT - ▪ Fuse failure ▪ Harness fault	▪ Check fuse ▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable rear spoiler supply fuse and circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the deployable rear spoiler valve circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C105A-19	Actuator Circuit Group A - Circuit current above threshold	**NOTE:** Circuit reference - VBATT - ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable rear spoiler power supply circuit for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the deployable rear spoiler valve circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105A-1D	Actuator Circuit Group A - Circuit current out of range	**NOTE:** Circuit reference - VBATT - ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable rear spoiler power supply circuit for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the deployable rear spoiler valve circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105A-64	Actuator Circuit Group A - Signal plausibility failure	**NOTE:** Circuit reference - VBATT - ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable rear spoiler power supply circuit for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the deployable rear spoiler valve circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105B-01	Actuator Circuit Group B - General Electrical Failure	**NOTE:** Circuit reference - VBATT - ▪ Fuse failure ▪ Harness fault	▪ Check fuse ▪ Refer to the electrical circuit diagrams and check the integrated suspension control module continuously variable damper power supply fuse and circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C105B-19	Actuator Circuit Group B - Circuit current above threshold	**NOTE:** Circuit reference - VBATT - ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module continuously variable damper power supply circuit for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105B-1D	Actuator Circuit Group B - Circuit current out of range	**NOTE:** Circuit reference - VBATT - ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module continuously variable damper power supply circuit for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C105B-64	Actuator Circuit Group B - Signal plausibility failure	**NOTE:** Circuit reference - VBATT - ▪ Harness fault	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module continuously variable damper power supply circuit for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the suspension damper solenoid circuits for short circuit to ground, short circuit to power, high resistance. Repair circuit as required, clear the DTCs and retest
C105D-1C	Deployable Rear Spoiler Deploy Relay Control - Circuit voltage out of range	**NOTE:** Circuit reference - SPOILER RAISE MONITOR - ▪ Deployable spoiler deployed monitoring circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	**NOTE:** Prior to initiating the diagnostic procedures described below, first check for DTC U3003-1C and note data logger records to ascertain vehicle speed when DTC C105D-1C was set. If vehicle speed was 200 kph or greater and if DTC U3003-1C was set during the same ignition cycle, then no action is required and DTC105D-1C may be cleared. Refer to the DTC index for DTC U3003-1C and rectify as required ▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable spoiler deployed monitoring circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required
C105D-72	Deployable Rear Spoiler Deploy Relay Control - Actuator stuck open	**NOTE:** Circuit reference - RAISE COIL NEGATIVE - ▪ Deployable spoiler deploy relay circuit open circuit, high resistance ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable spoiler deploy relay circuit open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required
C105D-73	Deployable Rear Spoiler Deploy Relay Control - Actuator stuck closed	**NOTE:** Circuit reference - RAISE COIL NEGATIVE - ▪ Deployable spoiler deploy relay circuit short circuit to ground ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable spoiler deploy relay circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C105E-1C	Deployable Rear Spoiler Retract Relay Control - Circuit voltage out of range	**NOTE:** Circuit reference - SPOILER LOWER MONITOR - ▪ Deployable spoiler retracted monitoring circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	**NOTE:** Prior to initiating the diagnostic procedures described below, first check for DTC U3003-1C and note data logger records to ascertain vehicle speed when DTC C105D-1C was set. If vehicle speed was 200 kph or greater and if DTC U3003-1C was set during the same ignition cycle, then no action is required and DTC105D-1C may be cleared. Refer to the DTC index for DTC U3003-1C and rectify as required ▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable spoiler retracted monitoring circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required
C105E-72	Deployable Rear Spoiler Retract Relay Control - Actuator stuck open	**NOTE:** Circuit reference - LOWER COIL NEG - ▪ Deployable spoiler retract relay circuit open circuit, high resistance ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable spoiler retract relay circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required
C105E-73	Deployable Rear Spoiler Retract Relay Control - Actuator stuck closed	**NOTE:** Circuit reference - LOWER COIL NEG - ▪ Deployable spoiler retract relay circuit short circuit to ground ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable spoiler retract relay circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C105F-72	Deployable Rear Spoiler Isolation Relay Control - Actuator stuck open	**NOTES:** ▪ Circuit reference - SPOILER POS - ▪ Circuit reference - SPOILER ISO RLY CTRL LSD - ▪ Deployable spoiler inhibit relay circuit between the integrated suspension control module and the passenger compartment junction box open circuit, high resistance ▪ Deployable spoiler inhibit relay ignition circuit between the JaguarDrive switchpack and the passenger compartment junction box open circuit, high resistance ▪ Deployable spoiler inhibit relay circuit between the central junction box and the passenger compartment junction box open circuit, high resistance ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	▪ Refer to the electrical circuit diagrams and check the integrated suspension control module deployable spoiler inhibit relay circuit between the integrated suspension control module and the passenger compartment junction box for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the deployable spoiler inhibit relay ignition circuit between the JaguarDrive switchpack and the passenger compartment junction box for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the deployable spoiler inhibit relay circuit between the central junction box and the passenger compartment junction box for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required
C105F-73	Deployable Rear Spoiler Isolation Relay Control - Actuator stuck closed	**NOTE:** Circuit reference - SPOILER ISO RLY CTRL LSD - ▪ Deployable spoiler inhibit relay circuit between the central junction box and the passenger compartment junction box short circuit to ground ▪ Relay fault (internal to passenger compartment junction box) ▪ Passenger compartment junction box failure	▪ Refer to the electrical circuit diagrams and check the deployable spoiler inhibit relay circuit between the central junction box and the passenger compartment junction box for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new passenger compartment junction box as required
C110C-01	Left Front Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Front left damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110C-12	Left Front Damper Solenoid - Circuit short to battery	▪ Front left damper solenoid circuit short circuit to power ▪ Front left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C110C-14	Left Front Damper Solenoid - Circuit short to ground or open	▪ Front left damper solenoid circuit short circuit to ground, open circuit, high resistance ▪ Front left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110C-19	Left Front Damper Solenoid - Circuit current above threshold	▪ Front left damper solenoid circuit short circuit between power and ground, high resistance ▪ Front left damper solenoid failure	**NOTE:** The resistance specification of the damper solenoid is 2 Ω to 3.5 Ω ▪ Refer to the electrical circuit diagrams and check the front left damper solenoid circuit for short circuit between power and ground, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110C-1D	Left Front Damper Solenoid - Circuit current out of range	▪ Front left damper solenoid circuit current out of range ▪ Front left damper solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110C-64	Left Front Damper Solenoid - Signal plausibility failure	▪ Front left damper solenoid signal plausibility failure ▪ Front left damper solenoid measured current control loop failed ▪ Front left damper solenoid circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check front left damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest ▪ Check front left damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C110D-01	Right Front Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Front right damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110D-12	Right Front Damper Solenoid - Circuit short to battery	▪ Front right damper solenoid circuit short circuit to power ▪ Front right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110D-14	Right Front Damper Solenoid - Circuit short to ground or open	▪ Front right damper solenoid circuit short circuit to ground, open circuit, high resistance ▪ Front right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C110D-18	Right Front Damper Solenoid - Circuit current above threshold	▪ Front right damper solenoid circuit short circuit between power and ground, high resistance ▪ Front right damper solenoid failure	**NOTE:** The resistance specification of the damper solenoid is 2 Ω to 3.5 Ω ▪ Refer to the electrical circuit diagrams and check the front right damper solenoid circuit for short circuit between power and ground, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110D-1D	Right Front Damper Solenoid - Circuit current out of range	▪ Front right damper solenoid circuit current out of range ▪ Front right damper solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110D-64	Right Front Damper Solenoid - Signal plausibility failure	▪ Front right damper solenoid signal plausibility failure ▪ Front right damper solenoid measured current control loop failed ▪ Front right damper solenoid circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check front right damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest ▪ Check front right damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C110E-01	Left Rear Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Rear left damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110E-12	Left Rear Damper Solenoid - Circuit short to battery	▪ Rear left damper solenoid circuit short circuit to power ▪ Rear left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110E-14	Left Rear Damper Solenoid - Circuit short to ground or open	▪ Rear left damper solenoid circuit short circuit to ground, open circuit, high resistance ▪ Rear left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110E-19	Left Rear Damper Solenoid - Circuit current above threshold	▪ Rear Left damper solenoid circuit short circuit between power and ground, high resistance ▪ Rear Left damper solenoid failure	**NOTE:** The resistance specification of the damper solenoid is 2 Ω to 3.5 Ω ▪ Refer to the electrical circuit diagrams and check the Rear Left damper solenoid circuit for short circuit between power and ground, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C110E-1D	Left Rear Damper Solenoid - Circuit current out of range	▪ Rear left damper solenoid circuit current out of range ▪ Rear left damper solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110E-64	Left Rear Damper Solenoid - Signal plausibility failure	▪ Rear left damper solenoid signal plausibility failure ▪ Rear left damper solenoid measured current control loop failed ▪ Rear left damper solenoid circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check rear left damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest ▪ Check rear left damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C110F-01	Right Rear Damper Solenoid - General electrical failure	▪ General electrical failure ▪ Rear right damper solenoid circuit fault	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110F-12	Right Rear Damper Solenoid - Circuit short to battery	▪ Rear right damper solenoid circuit short circuit to power ▪ Rear right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110F-14	Right Rear Damper Solenoid - Circuit short to ground or open	▪ Rear right damper solenoid circuit short circuit to ground, open circuit, high resistance ▪ Rear right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110F-19	Right Rear Damper Solenoid - Circuit current above threshold	▪ Rear Right damper solenoid circuit short circuit between power and ground, high resistance ▪ Rear Right damper solenoid failure	**NOTE:** The resistance specification of the damper solenoid is 2 Ω to 3.5 Ω ▪ Refer to the electrical circuit diagrams and check the Rear Right damper solenoid circuit for short circuit between power and ground, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required
C110F-1D	Right Rear Damper Solenoid - Circuit current out of range	▪ Rear right damper solenoid circuit current out of range ▪ Rear right damper solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear right damper solenoid failure	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new shock absorber as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C110F-64	Right Rear Damper Solenoid - Signal plausibility failure	▪ Rear right damper solenoid signal plausibility failure ▪ Rear right damper solenoid measured current control loop failed ▪ Rear right damper solenoid circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check rear right damper solenoid circuit resistance (nominal = 2 - 3.5 Ohms). Repair circuit as required, clear the DTCs and retest ▪ Check rear right damper solenoid circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1111-22	Control Lateral Acceleration - Signal amplitude > maximum	▪ Integrated suspension control module short circuit to power, short circuit to ground	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, contact the Jaguar Land Rover technical helpdesk for further guidance
C1124-56	Height Sensor(s) - Invalid / Incomplete configuration	▪ Height sensor(s) not calibrated	▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Height Sensor Calibration. Clear the DTCs and retest
C112F-72	Air Spring Valve - Actuator stuck open	▪ Corner valve stuck open (fully or partially) ▪ Vehicle driven while system in "Tight Tolerance" mode	▪ Complete corner valve checks. If necessary, clear tight tolerance mode. Clear the DTC and retest
C1130-53	Air Spring Air Supply - Deactivated	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to deflate ▪ Diagnostic routine activity only	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to deflate ▪ Ignore/clear this DTC

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1130-66	Air Spring Air Supply - Signal has too many transitions/events	▪ Leaking air spring or pipe to air spring ▪ Loose pipe fitting ▪ Corner valve leak to gallery ▪ Air suspension air supply unit fault - Relief pressure too low	▪ Check that there is pressure in the air spring. If the air spring is not under pressure, then use the Jaguar Land Rover Approved Diagnostic Equipment to open the corner valve and run the suspension air supply unit. Stop the suspension air supply unit if the gallery pressure exceeds 900 Kpa. Use leak detection spray on the connections to the air spring and the axle valve block to check for leaks while the air spring is being pressurised. If a leak is detected in the air spring unit, replace the air spring damper assembly as required. If a leak is detected in a connection, repair or replace the connector as required. Clear the DTCs and retest ▪ If no leak is evident, then inspect the air spring pipe and the associated axle valve block. If a leak is detected, rectify using an approved repair kit or replace the faulty pipe as necessary. Clear the DTCs and retest ▪ To check for a leaking corner valve, use the Jaguar Land Rover Approved Diagnostic Equipment to open the exhaust valve, then close all valves and monitor gallery pressure to see if it continues to increases after 2 minutes. If the gallery pressure continues to rise then it is likely that a corner valve or the reservoir is leaking. Disconnect the front gallery pipe from the rear valve block, fit a tight fitting hose over the end of this pipe and place the other end in a container of water. If a continuous flow of bubbles is observed then a front corner valve is leaking and the front valve block should be replaced. If no bubbles are observed then fit a temporary pipe to the front gallery port of the rear valve block and place the other end in a container of water. If a continuous flow of bubbles is observed then a rear corner valve or the reservoir valve may be leaking and the rear valve block should be replaced. Clear the DTCs and retest ▪ Use the Jaguar Land Rover Approved Diagnostic Equipment to vent the common gallery through the exhaust valve. Then disconnect the 8mm delivery pipe from the delivery port on the back of the air dryer. Connect a calibrated pressure gauge tool directly into the delivery port. (Note that an 8mm connector will be required). Run the suspension air supply unit and check that the unit can generate a minimum of 18 bar air pressure. If 18 bar or greater can be generated, then the unit is operating correctly. Remove the pressure gauge and make sure that all pipe work is re-installed correctly ▪ If pressure is lower than 18 bar, disconnect the suspension air supply unit intake pipe and run the suspension air supply unit again. If 18 bar pressure or greater can be generated with the intake pipe disconnected, then check for obstructions/blockages in the intake pipes and filter. Repair or replace intake system as required ▪ If a minimum of 18 bar pressure cannot be generated with the intake pipe disconnected, check and install a new air suspension air supply unit as required. Clear DTCs and retest
C1130-7A	Air Spring Air Supply - Fluid leak or seal failure	▪ Detached or burst air pipe ▪ Leaking air spring or pipe to air spring ▪ Loose or leaking air pipe connection	▪ Check that there is pressure in the air spring. If the air spring is not under pressure, then use the Jaguar Land Rover Approved Diagnostic Equipment to open the corner valve and run the suspension air supply unit. Stop the suspension air supply unit if the gallery pressure exceeds 900 Kpa. Use leak detection spray on the connections to the air spring and the axle valve block to check for leaks while the air spring is being pressurised. If a leak is detected in the air spring unit, replace the air spring damper assembly as required. If a leak is detected in a connection, repair or replace the connector as required. Clear DTCs and retest ▪ If no leak is evident, then inspect the air spring pipes and the associated axle valve blocks. If a leak is detected, rectify using an approved repair kit or replace the faulty pipe as necessary. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1131-92	Air Supply - Performance or incorrect operation	• Pressure sensor fault • Reservoir valve fault • Exhaust valve fault • Suspension air supply unit fault • Gallery or reservoir pipe kinked, crushed, blocked or leaking	• Using the Jaguar Land Rover approved diagnostic equipment, check for pressure sensor related DTCs and perform the relevant corrective actions • Using the Jaguar Land Rover approved diagnostic equipment, check for reservoir valve related DTCs and perform the relevant corrective actions • Using the Jaguar Land Rover approved diagnostic equipment, check for exhaust valve related DTCs and perform the relevant corrective actions • Using the Jaguar Land Rover approved diagnostic equipment, check for suspension air supply unit related DTCs and perform the relevant corrective actions • Check the reservoir pipe and gallery pipe for kinks, crushing, blockages and leaks. Rectify as necessary
C1138-92	Access Switch - Performance or incorrect operation	• Vehicle access switch (switch on driver's door) circuit - Short circuit to ground • Driver's door switch pack fault	• Check the operation of the vehicle access switch and repair/renew as necessary • Refer to the electrical circuit diagrams and check the vehicle access switch circuit from the driver's door to the integrated suspension control module for short circuit to ground. Repair circuit as required, clear the DTCs and retest • If the fault persists, install a new driver's door switch pack. Clear the DTCs and retest
C1A03-14	Left Front Height Sensor - Circuit short to ground or open	• Height sensor connector loose • Front left height sensor circuit short circuit to ground, open circuit, high resistance • Front left height sensor circuit short circuit to another circuit • Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated • To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below • Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required • Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only • Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest • If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A03-17	Left Front Height Sensor - Circuit voltage above threshold	▪ Front left height sensor circuit short circuit to another circuit ▪ Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A03-21	Left Front Height Sensor - Signal amplitude < minimum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Front left height sensor circuit short circuit to ground ▪ Front left height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A03-22	Left Front Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Front left height sensor circuit short circuit to ground ▪ Front left height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure
C1A03-26	Left Front Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Front left height sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, refer to the electrical circuit diagrams and check left front height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A03-27	Left Front Height Sensor - Signal rate of change above threshold	▪ Height sensor fault ▪ Front left height sensor circuit short circuit to ground, short circuit to power, intermittent open circuit, high resistance	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, refer to the electrical circuit diagrams and check left front height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A03-64	Left Front Height Sensor - Signal plausibility failure	▪ Front left height sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front left height sensor mechanical failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Refer to the electrical circuit diagrams and check the front left height sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Check the integrity of the front left height sensor and linkage. Rectify as required
C1A03-92	Left Front Height Sensor - Performance or incorrect operation	▪ Front left height sensor performance not as expected/signal changing slower than expected ▪ Height sensor incorrectly installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height Using the Jaguar Land Rover approved diagnostic equipment. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A04-14	Right Front Height Sensor - Circuit short to ground or open	▪ Height sensor connector loose ▪ Front right height sensor circuit short circuit to ground, open circuit, high resistance ▪ Front right height sensor circuit short circuit to another circuit ▪ Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest
C1A04-17	Right Front Height Sensor - Circuit voltage above threshold	▪ Front right height sensor circuit short circuit to another circuit ▪ Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear DTC and retest ▪ If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A04-21	Right Front Height Sensor - Signal amplitude < minimum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Front right height sensor circuit short circuit to ground ▪ Front right height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A04-22	Right Front Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Front right height sensor circuit short circuit to ground ▪ Front right height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure
C1A04-26	Right Front Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Front right height sensor circuit - Short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, refer to the electrical circuit diagrams and check front right height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A04-27	Right Front Height Sensor - Signal rate of change above threshold	▪ Height sensor fault ▪ Front right height sensor circuit short circuit to ground, short circuit to power, intermittent open circuit, high resistance	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, refer to the electrical circuit diagrams and check right front height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A04-64	Right Front Height Sensor - Signal plausibility failure	▪ Front right height sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Front right height sensor mechanical failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Refer to the electrical circuit diagrams and check the front right height sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Check the integrity of the front right height sensor and linkage. Rectify as required
C1A04-92	Right Front Height Sensor - Performance or incorrect operation	▪ Front right height sensor performance not as expected/signal changing slower than expected ▪ Height sensor incorrectly installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height Using the Jaguar Land Rover approved diagnostic equipment. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A05-14	Left Rear Height Sensor - Circuit short to ground or open	▪ Height sensor connector loose ▪ Rear left height sensor circuit - Short circuit to ground, open circuit, high resistance ▪ Rear left height sensor circuit short circuit to another circuit ▪ Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest
C1A05-17	Left Rear Height Sensor - Circuit voltage above threshold	▪ Rear left height sensor circuit short circuit to another circuit ▪ Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear DTC and retest ▪ If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A05-21	Left Rear Height Sensor - Signal amplitude < minimum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear left height sensor circuit short circuit to ground ▪ Rear left height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A05-22	Left Rear Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear left height sensor circuit short circuit to ground ▪ Rear left height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure
C1A05-26	Left Rear Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Rear left height sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, refer to the electrical circuit diagrams and check rear left height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A05-27	Left Rear Height Sensor - Signal rate of change above threshold	▪ Height sensor fault ▪ Rear left height sensor circuit short circuit to ground, short circuit to power, intermittent open circuit, high resistance	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, refer to the electrical circuit diagrams and check rear left height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A05-64	Left Rear Height Sensor - Signal plausibility failure	▪ Rear left height sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear left height sensor mechanical failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Refer to the electrical circuit diagrams and check the rear left height sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Check the integrity of the rear left height sensor and linkage. Rectify as required
C1A05-92	Left Rear Height Sensor - Performance or incorrect operation	▪ Rear left height sensor performance not as expected/signal changing slower than expected ▪ Height sensor incorrectly installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height Using the Jaguar Land Rover approved diagnostic equipment. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A06-14	Right Rear Height Sensor - Circuit short to ground or open	▪ Height sensor connector loose ▪ Rear right height sensor circuit short circuit to ground, open circuit, high resistance ▪ Rear right height sensor circuit short circuit to another circuit ▪ Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear DTC and retest ▪ If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest
C1A06-17	Right Rear Height Sensor - Circuit voltage above threshold	▪ Rear right height sensor circuit short circuit to another circuit ▪ Height sensor failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear DTC and retest ▪ If the fault persists, check and install a new height sensor as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A06-21	Right Rear Height Sensor - Signal amplitude < minimum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear right height sensor circuit short circuit to ground ▪ Rear right height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A06-22	Right Rear Height Sensor - Signal amplitude > maximum	▪ Height sensor linkage not connected ▪ Height sensor or bracket loose ▪ Height sensor bracket bent ▪ Incorrect height calibration ▪ Height sensor linkage toggled ▪ Rear right height sensor circuit short circuit to ground ▪ Rear right height sensor circuit short circuit to another circuit ▪ Height sensor electrical fault ▪ Incorrect height sensor installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated ▪ Inspect height sensor for damage or loose fixings. Confirm that the correct height sensor part number is installed, as specified in the service parts database ▪ To check height sensor: disconnect electrical connector to height sensor and inspect connector pins and terminals for evidence of corrosion or water ingress. If no corrosion found, disconnect harness at control module and follow the three steps below ▪ Check for short circuits between any of the three terminals and vehicle ground. Repair circuit as required ▪ Check for electrical continuity between the two connectors for each of the three terminals. Repair circuit as required. Reconnect electrical connector at control module only ▪ Check voltages at terminals within height sensor connector (with sensor not connected): voltage to sensor ground connection should be ~0V, voltage to sensor signal connection should be ~0V, voltage to sensor supply connection should be ~5V. All voltages should be within ± 0.15V. Repair circuit as required, clear the DTCs and retest ▪ To check sensor operation on the vehicle: Check for water ingress around the height sensors, electrical connectors or shaft end. Check for excessive movement in the shaft in all directions. Raise vehicle (ideally on wheels-free ramp) until suspension on corner under investigation is at full extension to gain access to height sensor. Access may be improved by removing road-wheel. Carefully disconnect the height sensor link from the upper suspension arm. Monitor the height sensor signal voltage output for the height sensor under investigation. Position the sensor arm so it is in the mid position and confirm that the voltage is around 2.5 volts. Move the sensor arm over the range ±40° around the mid position and confirm that the voltage changes smoothly between around 0.2 volts and 4.8 volts. If voltages are incorrect or do not change smoothly then renew the sensor. NOTE: For angles of movement beyond ±40°, the sensor signal will remain at a voltage of ~0.15V or ~4.85V, depending on position of sensor lever. This is normal. When investigation is complete, install the height sensor link to the upper arm. If any fixings to the height sensor body or mounting bracket were slackened, found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated. Refer to the relevant section of the workshop manual for the calibration procedure
C1A06-26	Right Rear Height Sensor - Signal rate of change below threshold	▪ Height sensor linkage disconnected ▪ Height sensor fault ▪ Rear right height sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest ▪ If the fault persists, refer to the electrical circuit diagrams and check rear right height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A06-27	Right Rear Height Sensor - Signal rate of change above threshold	• Height sensor fault • Rear right height sensor circuit short circuit to ground, short circuit to power, intermittent open circuit, high resistance	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated • Check height sensor linkage is securely connected and not damaged. Visually inspect the height sensor for security, damage and correct orientation/installation. Check the sensor connector for damage. Repair/renew as required, clear the DTCs and retest • If the fault persists, refer to the electrical circuit diagrams and check rear right height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest
C1A06-64	Right Rear Height Sensor - Signal plausibility failure	• Rear right height sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance • Rear right height sensor mechanical failure	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated • Refer to the electrical circuit diagrams and check the rear right height sensor signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest • Check the integrity of the rear right height sensor and linkage. Rectify as required
C1A06-92	Right Rear Height Sensor - Performance or incorrect operation	• Rear right height sensor performance not as expected/signal changing slower than expected • Height sensor incorrectly installed	**NOTE:** If any height sensor fixings were slackened or found to be loose or if a height sensor was changed, the vehicle ride height MUST be re-calibrated • Check that the vehicle is free of obstructions. Check the height sensor for correct installation and torque of fixings. If any fixings require attention, re-calibrate the ride height Using the Jaguar Land Rover approved diagnostic equipment. Clear the DTCs and retest
C1A24-67	No Temperature Increase When Compressor Requested - Signal incorrect after event	• Signal plausibility failure • Air suspension air supply unit cylinder head temperature sensor - Circuit fault • Air suspension air supply unit cylinder head temperature sensor not attached to the suspension air supply unit • Air suspension air supply unit cylinder head temperature sensor - Internal failure	• Use the Jaguar Land Rover Approved Diagnostic Equipment to check if the suspension air supply unit will operate. If the unit does not operate check the air suspension air supply unit fuses and relays and check the power and ground circuits for open circuit, high resistance. Repair circuit(s) as required, clear the DTCs and retest • If the unit will operate through the Jaguar Land Rover Approved Diagnostic Equipment, disconnect the control module connector and check the resistance between the terminals on the harness connector using a digital multimeter. If the ambient temperature is >0°C and the resistance exceeds 6 MegaOhms then the wiring and the thermistor should be investigated. Check the resistance between the relevant pins of the suspension air supply unit four-way connector. Repair circuit(s) as required, clear the DTCs and retest • If the fault persists, install a new air suspension air supply unit, clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A36-11	Exhaust Valve - Circuit short to ground	▪ Exhaust valve circuit - Short circuit to ground ▪ Exhaust valve failure	▪ Refer to the electrical circuit diagrams and check the exhaust valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the exhaust valve solenoid for short circuit to ground. If valve fault exists, check and install a new suspension air supply unit as required. Clear the DTCs and retest
C1A36-12	Exhaust Valve - Circuit short to battery	▪ Exhaust valve circuit - Short circuit to power ▪ Exhaust valve failure	▪ Refer to the electrical circuit diagrams and check the exhaust valve circuit for short circuit to power. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the exhaust valve solenoid for short circuit to power. If valve fault exists, check and install a new suspension air supply unit as required. Clear the DTCs and retest
C1A36-13	Exhaust Valve - Circuit open	▪ Exhaust valve circuit - Open circuit, high resistance ▪ Exhaust valve disconnected ▪ Exhaust valve failure	▪ Refer to the electrical circuit diagrams and check the exhaust valve circuit for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check integrity of rear valve block connector. Repair or replace as required. Clear the DTCs and retest ▪ If the fault persists, install a new solenoid valve. Clear the DTCs and retest
C1A37-11	Front Cross-Link Valve - Circuit short to ground	▪ Front cross-link valve circuit - Short circuit to ground ▪ Front cross-link valve failure	▪ Refer to the electrical circuit diagrams and check the front cross-link valve circuit for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the front cross-link valve solenoid for short circuit to ground. If valve fault exists, check and install a new front valve block assembly as required. Clear the DTCs and retest
C1A68-1C	Left Front Height Sensor Supply - Circuit voltage out of range	▪ Front left height sensor supply circuit voltage out of range ▪ Height sensor wiring harness short circuit to ground, short circuit to power, open circuit, high resistance ▪ Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated ▪ Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system ▪ Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new height sensor. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1A69-1C	Right Front Height Sensor Supply - Circuit voltage out of range	▪ Front right height sensor supply circuit voltage out of range ▪ Height sensor wiring harness short circuit to ground, short circuit to power, open circuit, high resistance ▪ Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated ▪ Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system ▪ Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new height sensor. Clear the DTCs and retest
C1A70-1C	Left Rear Height Sensor Supply - Circuit voltage out of range	▪ Rear left height sensor supply circuit voltage out of range ▪ Height sensor wiring harness short circuit to ground, short circuit to power, open circuit, high resistance ▪ Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated ▪ Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system ▪ Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new height sensor. Clear the DTCs and retest
C1A71-1C	Right Rear Height Sensor Supply - Circuit voltage out of range	▪ Rear right height sensor supply circuit voltage out of range ▪ Height sensor wiring harness short circuit to ground, short circuit to power, open circuit, high resistance ▪ Height sensor failure	**NOTE:** If a height sensor is changed, the vehicle ride height must be re-calibrated ▪ Where available, refer to the guided diagnostic routine for this DTC on the approved diagnostic system ▪ Check the height sensor connector for damage and security. Refer to the electrical circuit diagrams and check height sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new height sensor. Clear the DTCs and retest
C1B15-11	Sensor Supply Voltage B - Circuit short to ground	▪ Vertical accelerometer supply circuit(s) short circuit to ground ▪ Vertical accelerometer failure	▪ Refer to the electrical circuit diagrams and check the vertical accelerometer supply circuit(s) for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install new vertical accelerometer(s) as required. After installation, calibrate the accelerometer/integrated suspension control module, as required. Clear the DTCs and retest ▪ If the fault persists, check control module sensor supply output voltage. Measured voltage should be between 4.75 volts and 4.96 volts. If output voltage is out of range, check and install a new integrated suspension control module as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C1B15-17	Sensor Supply Voltage B - Circuit voltage above threshold	▪ Vertical accelerometer supply circuit(s) short circuit to power, short circuit to another circuit ▪ Vertical accelerometer failure	▪ Refer to the electrical circuit diagrams and check the vertical accelerometer supply circuit(s) for short circuit to power, short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install new vertical accelerometer(s) as required. After installation, calibrate the accelerometer/integrated suspension control module, as required. Clear the DTCs and retest ▪ If the fault persists, check control module sensor supply output voltage. Measured voltage should be between 4.75 volts and 4.96 volts. If output voltage is out of range, check and install a new integrated suspension control module as required. Clear the DTCs and retest
C1B15-1C	Sensor Supply Voltage B - Circuit voltage out of range	▪ Vertical accelerometer supply circuit(s) short circuit to ground, short circuit to another circuit ▪ Vertical accelerometer failure	▪ Refer to the electrical circuit diagrams and check the vertical accelerometer supply circuit(s) for short circuit to ground, short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install new vertical accelerometer(s) as required. After installation, calibrate the accelerometer/integrated suspension control module, as required. Clear the DTCs and retest ▪ If the fault persists, check control module sensor supply output voltage. Measured voltage should be between 4.75 volts and 4.96 volts. If output voltage is out of range, check and install a new integrated suspension control module as required. Clear the DTCs and retest
C1B18-2F	Module Power Supplies - Signal erratic	▪ Loss of power to the control module when driving	▪ Check the module connector(s) for security and integrity. Rectify as required ▪ Refer to the electrical circuit diagrams and check the power supply circuit(s) to the module for poor or intermittent connection. Repair circuit(s) as required, clear the DTCs and retest
C1B18-62	Module Power Supplies - Signal compare failure	▪ Integrated suspension control module unexpected power fluctuations detected ▪ Integrated suspension control module failure	▪ Check the module connector(s) for security and integrity. Rectify as required ▪ Refer to the electrical circuit diagrams and check the power supply and ground circuit(s) to the module for poor or intermittent connection. Repair circuit(s) as required, clear the DTCs and retest ▪ If the fault persists, check and install a new integrated suspension control module as required
C1B18-94	Module Power Supplies - Unexpected operation	▪ Integrated suspension control module cannot power down as required ▪ Integrated suspension control module failure	▪ Clear the DTCs and retest. If the fault persists, check and install a new integrated suspension control module as required
P1905-00	Control Module Configured for End-of-Line Test Mode - No sub type information	▪ Integrated suspension control module not correctly configured	▪ Using the Jaguar Land Rover approved diagnostic equipment, configure the module as a new module. Clear the DTCs and retest
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module through the high speed CAN bus (chassis and powertrain)	**NOTE:** Do NOT install a new integrated suspension control module as this will NOT rectify the fault ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	▪ Invalid data received from another control module through the high speed CAN bus (chassis and powertrain)	**NOTE:** Do NOT install a new integrated suspension control module as this will NOT rectify the fault ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-83	High Speed CAN Communication Bus - Value of signal protection calculation incorrect	▪ Invalid data received from another control module through the high speed CAN bus (chassis and powertrain)	**NOTE:** Do NOT install a new integrated suspension control module as this will NOT rectify the fault ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-86	High Speed CAN Communication Bus - Signal invalid	▪ The integrated suspension control module has learned a steering sensor offset value of greater than +/-10 degrees over time ▪ Steering angle sensor offset out of range ▪ Steering angle sensor faulty ▪ Steering angle sensor incorrectly installed ▪ Wheels out of alignment. Incorrect steering offset data learnt by integrated suspension control module	**NOTE:** Do NOT install a new integrated suspension control module as this will NOT rectify the fault ▪ Check steering angle sensor alignment ▪ Check wheel alignment ▪ Check steering sensor for correct operation ▪ Check steering sensor offset PID 3B4B, if >+/-10 degrees then clear integrated suspension control module adaptive data
U0001-87	High Speed CAN Communication Bus - Missing message	▪ Missing message from another control module through the high speed CAN bus (chassis and powertrain)	**NOTE:** Do NOT install a new integrated suspension control module as this will NOT rectify the fault ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the relevant control module (i.e. as indicated by the snapshot data) power and ground circuits for open circuit. Repair circuits as required, clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN network (chassis and powertrain) between the relevant control module (i.e. as indicated by the snapshot data) and the integrated suspension control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0028-88	Vehicle Communication Bus A - Bus Of	▪ High speed CAN (powertrain) fault	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN network (powertrain)
U0046-88	Vehicle Communication Bus C - Bus Off	▪ High speed CAN (chassis) fault	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN network (chassis)
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ Car configuration signal not received ▪ Car configuration file incorrect ▪ The integrated suspension control module is not compatible with the vehicle	▪ Using the Jaguar Land Rover approved diagnostic equipment check/amend the car configuration file. Clear the DTCs and retest ▪ If the fault persists, check the integrated suspension control module part number, install the correct part as required. Clear the DTCs and retest
U0300-57	Internal Control Module Software Incompatibility - Invalid / incomplete software component	▪ Incorrect software for the integrated suspension control module has been installed	▪ Using the Jaguar Land Rover approved diagnostic equipment check/amend the module software. Clear the DTCs and retest
U1A14-00	CAN Initialization Failure - No sub type information	▪ CAN network harness short circuit or disconnected	▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network ▪ Refer to the electrical circuit diagrams and check the CAN network between the central junction box and the integrated suspension control module
U2011-11	Motor - Circuit short to ground	**NOTES:** ▪ Circuit reference - MOTOR DOWN - ▪ Circuit reference - MOTOR UP - ▪ Deployable spoiler assembly motor circuit short circuit to ground ▪ Deployable spoiler assembly failure	▪ Refer to the electrical circuit diagrams and check the deployable spoiler assembly motor circuit between the deployable spoiler and the passenger compartment junction box for short circuit to ground. Repair circuit as required, clear the DTCs and retest ▪ Refer the electrical circuit diagrams and check for short circuit to ground at deployable spoiler motor down and motor up terminals. Repair circuit as required, clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, run the deployable rear spoiler self test diagnostic routine ▪ If the fault persists, check and install a new deployable spoiler assembly as required. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2011-13	Motor - Circuit open	**NOTES:** ▪ Circuit reference - MOTOR DOWN - ▪ Circuit reference - MOTOR UP - ▪ Deployable spoiler assembly motor circuit open circuit, high resistance ▪ Deployable spoiler assembly failure	▪ Refer to the electrical circuit diagrams and check the deployable spoiler assembly motor circuit between the deployable spoiler and the passenger compartment junction box for open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check for open circuit, high resistance at deployable spoiler motor down and motor up terminals. Repair circuit as required, clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, run the deployable rear spoiler self test diagnostic routine ▪ If the fault persists, check and install a new deployable spoiler assembly as required. Clear the DTCs and retest
U2011-71	Motor - Actuator stuck	**NOTES:** ▪ Circuit reference - LOWERED SIG - ▪ Circuit reference - RAISE SIG - ▪ Deployable spoiler assembly signal circuit(s) between the deployable spoiler and the integrated suspension control module short circuit to ground, short circuit to power, open circuit, high resistance ▪ Switch fault (internal to deployable spoiler assembly) ▪ Deployable spoiler assembly failure	▪ Refer to the electrical circuit diagrams and check the deployable spoiler assembly signal circuit(s) between the deployable spoiler and the integrated suspension control module for short circuit to ground, short circuit to power, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, run the deployable rear spoiler self test diagnostic routine ▪ If the fault persists, check and install a new deployable spoiler assembly as required. Clear the DTCs and retest
U201A-54	Control Module Main Calibration Data - Missing calibration	▪ A new integrated suspension control module has been installed and is not correctly configured	▪ Using the Jaguar Land Rover approved diagnostic equipment, configure the module as a new module. Clear the DTCs and retest
U201A-57	Control Module Main Calibration Data - Invalid / incomplete software component	▪ A new integrated suspension control module has been installed and is not correctly configured ▪ Incorrect calibration data has been installed ▪ Incorrect software has been installed	▪ Using the Jaguar Land Rover approved diagnostic equipment, confirm correct calibration data and software versions and install as required. Clear the DTCs and retest
U2100-00	Initial Configuration Not Complete - No sub type information	▪ Car configuration data has not been loaded correctly ▪ New central junction box has been installed but not initialized ▪ Central junction box fault	▪ Check/amend the car configuration file Using the Jaguar Land Rover approved diagnostic equipment. Clear the DTCs and retest ▪ Check the central junction box for related DTCs and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2101-00	Control Module Configuration Incompatible - No sub type information	▪ Car configuration data transmitted over CAN does not match integrated suspension control module internal configuration	▪ Complete the new module software installation procedure Using the Jaguar Land Rover approved diagnostic equipment. Clear the DTCs and retest
U210A-16	Temperature Sensor - Circuit voltage below threshold	▪ Air suspension air supply unit cylinder head temperature sensor circuit - Circuit voltage below threshold ▪ Air suspension air supply unit cylinder head temperature sensor circuit - Short circuit to ground or short circuit to another circuit	▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit cylinder head temperature sensor circuit between the suspension air supply unit and the integrated suspension control module for short circuit to ground or short circuit to another circuit. Repair circuit as required, clear the DTCs and retest ▪ Refer to the electrical circuit diagrams and check the air suspension air supply unit cylinder head temperature sensor circuit within the air suspension air supply unit. Check for wiring damage within the air suspension air supply unit. Repair circuit or replace air suspension air supply unit harness as required, clear the DTCs and retest. ▪ If the fault persists, install a new air suspension air supply unit
U2300-54	Central Configuration - Missing calibration	▪ Car configuration file mismatch with vehicle specification ▪ Integrated suspension control module is not configured correctly	**NOTE:** After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, re-configure the integrated suspension control module with the latest level software. Clear the DTCs and retest
U2300-55	Central Configuration - Not configured	▪ Car configuration file mismatch with vehicle specification ▪ Integrated suspension control module is not configured correctly	**NOTE:** After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, re-configure the integrated suspension control module with the latest level software. Clear the DTCs and retest
U2300-56	Central Configuration - Invalid / Incomplete configuration	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, re-configure the integrated suspension control module with the latest level software. Clear the DTCs and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2300-64	Central Configuration - Signal plausibility failure	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 40 seconds before clearing the DTCs ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the DTCs and retest ▪ Using the Jaguar Land Rover approved diagnostic equipment, re-configure the integrated suspension control module with the latest level software. Clear the DTCs and retest
U3000-04	Control Module - System internal failures	▪ Control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, contact the Jaguar Land Rover technical helpdesk for further guidance
U3000-05	Control Module - System programming failures	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to operate in tight tolerance mode ▪ The vehicle suspension is in tight tolerance mode (transit mode)	**NOTE:** This DTC does not represent a fault and is triggered by routine diagnostic activity when the air suspension is requested to operate in tight tolerance mode ▪ Connect a Jaguar Land Rover approved Midtronics battery power supply to the vehicle startup battery ▪ Connect the Jaguar Land Rover approved diagnostic equipment to the vehicle and begin a new session. The diagnostic equipment will detect the correct Vehicle Identification Number (VIN) for the current vehicle and automatically take the vehicle out of transit mode if required ▪ Following all the on-screen instructions, select 'ECU Diagnostics'; then, select 'Chassis Control Module [CHCM]'; then, select 'ECU Functions'; then, select 'Suspension geometry set-up' and, when prompted, select 'Normal Mode'. Continue to follow all the on-screen instructions until the application completes successfully. Finally, exit the current diagnostic session ▪ Disconnect the diagnostic equipment and battery power supply from the vehicle. Clear the DTCs and retest to confirm the DTC is no longer present and any warning messages that may have been displayed on the instrument cluster are no longer showing
U3000-43	Control Module - Special memory failure	▪ Control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new integrated suspension control module as required
U3000-45	Control Module - Program memory failure	▪ Control module failure	▪ Refer to the electrical circuit diagrams and check the power and ground circuits to the module. Repair circuit as required, clear the DTCs and retest ▪ If the fault persists, check and install a new integrated suspension control module as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3000-47	Control Module - Watchdog/safety microcontroller failure	▪ Control module - Internal watchdog/safety microcontroller failure ▪ The control module has completed a software reset	▪ Clear the DTCs and retest. If the fault persists, contact retailer technical support
U3000-54	Control Module - Missing calibration	▪ The control module has been replaced and has not been calibrated	▪ Using the Jaguar Land Rover approved diagnostic equipment, calibrate the vehicle ride height. Clear the DTCs and retest
U3003-1C	Battery Voltage - Circuit voltage out of range	▪ Power supply circuit - Voltage out of range (supply voltage at control module below 10.5V or above 18V for 30 seconds)	▪ Check the battery is in good condition and fully charged, refer to the battery care requirements ▪ Refer to the starting and charging section of the workshop manual and check the performance of the charging system ▪ Refer to the electrical circuit diagrams and check power and ground circuit to the control module for faults, including intermittent high resistance. Repair circuit as required, clear the DTCs and retest
U3003-62	Battery Voltage - Signal compare failure	▪ Signal compare failure ▪ High resistance connections	▪ Check the battery is in good condition and fully charged, refer to the battery care requirements ▪ Refer to the starting and charging section of the workshop manual and check the performance of the charging system ▪ Refer to the electrical circuit diagrams and check power and ground circuit to the control module for faults, including intermittent high resistance. Repair circuit as required, clear the DTCs and retest
U300A-1C	Ignition Switch - Circuit voltage out of range	▪ Switched ignition supply circuit voltage out of range	▪ Refer to the electrical circuit diagrams and check the switched ignition supply circuit to the control module for faults, including intermittent high resistance. Repair circuit as required, clear the DTCs and retest
U300A-62	Ignition Switch - Signal compare failure	▪ Mismatch between hardwired ignition voltage supply and control module internal measured voltage	▪ Check the security and integrity of the switched ignition supply circuit connectors. Rectify as required ▪ Refer to the electrical circuit diagrams and check the switched ignition supply circuit to the control module for faults, including intermittent high resistance. Repair circuit as required, clear the DTCs and retest
U300A-64	Ignition Switch - Signal plausibility failure	▪ Switched ignition supply circuit short circuit to ground, open circuit, high resistance	▪ Check the security and integrity of the switched ignition supply circuit connectors. Rectify as required ▪ Refer to the electrical circuit diagrams and check the switched ignition supply circuit to the control module for short circuit to ground, open circuit, high resistance. Repair circuit as required, clear the DTCs and retest

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